

AIRPORT CARGO OPERATIONS

Predictive Analytics Report

Modeling & Forecasting Shipment Processing Time

Data Cleaning • Exploratory Analysis • Feature Engineering • Regression Modeling

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1. Executive Summary

This report documents an end-to-end analytics workflow applied to airport cargo shipment data, covering data cleaning, exploratory data analysis (EDA), feature engineering, and predictive modeling. The objective was to understand the operational drivers of cargo processing time and to build a regression model capable of forecasting it from shipment characteristics.

The dataset comprises 250 shipment records with attributes covering weight, priority level, destination zone, warehouse distance, and the number of handling agents assigned. After resolving missing values and formatting inconsistencies, a Linear Regression model was trained to predict Processing Time (in minutes). The model explained 93.4% of the variance in processing time ($R^2 = 0.9342$) with a root mean squared error of 10.78 minutes, indicating strong predictive performance.

93.4% R ² SCORE (VARIANCE EXPLAINED)	10.78 RMSE (MINUTES)	250 SHIPMENT RECORDS ANALYZED
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2. Dataset Overview

The source dataset, `airport_cargo_regression_practice.xlsx`, contains 250 shipment records across seven fields. The table below summarizes each field and its role in the analysis.

Column	Type	Description	Role
Shipment_ID	Integer	Unique shipment identifier	Identifier (dropped before modeling)
Weight_KG	Float	Cargo weight in kilograms	Numerical feature
Priority_Level	Categorical	Low / Medium / High	Ordinal feature
Destination_Zone	Categorical	Zone_A – Zone_D	Nominal feature
Warehouse_Distance_Meters	Float	Distance from warehouse to gate	Numerical feature
Handling_Agents	Integer	Agents assigned to the shipment	Numerical feature
Processing_Time_Mins	Float	Total processing time	Target variable

Baseline descriptive statistics (post-cleaning) for the numerical fields:

Statistic	Weight_KG	Warehouse_Distance_M	Handling_Agents	Processing_Time_Mins
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Statistic	Weight_KG	Warehouse_Distance_M	Handling_Agents	Processing_Time_Mins
Mean	1273.49	407.80	3.00	110.39
Std. Dev.	694.61	214.60	1.44	46.34
Min	112.10	53.50	1	15.00
Median	1299.85	400.30	3	116.75
Max	2468.50	797.70	5	207.30

3. Data Cleaning

Initial inspection surfaced two data-quality issues that required remediation before analysis:

- Missing values: the Weight_KG column contained 12 null entries (4.8% of records).
- Formatting inconsistency: numeric values in Weight_KG contained thousands-separator commas, causing the column to be misread as text.

Cleaning approach

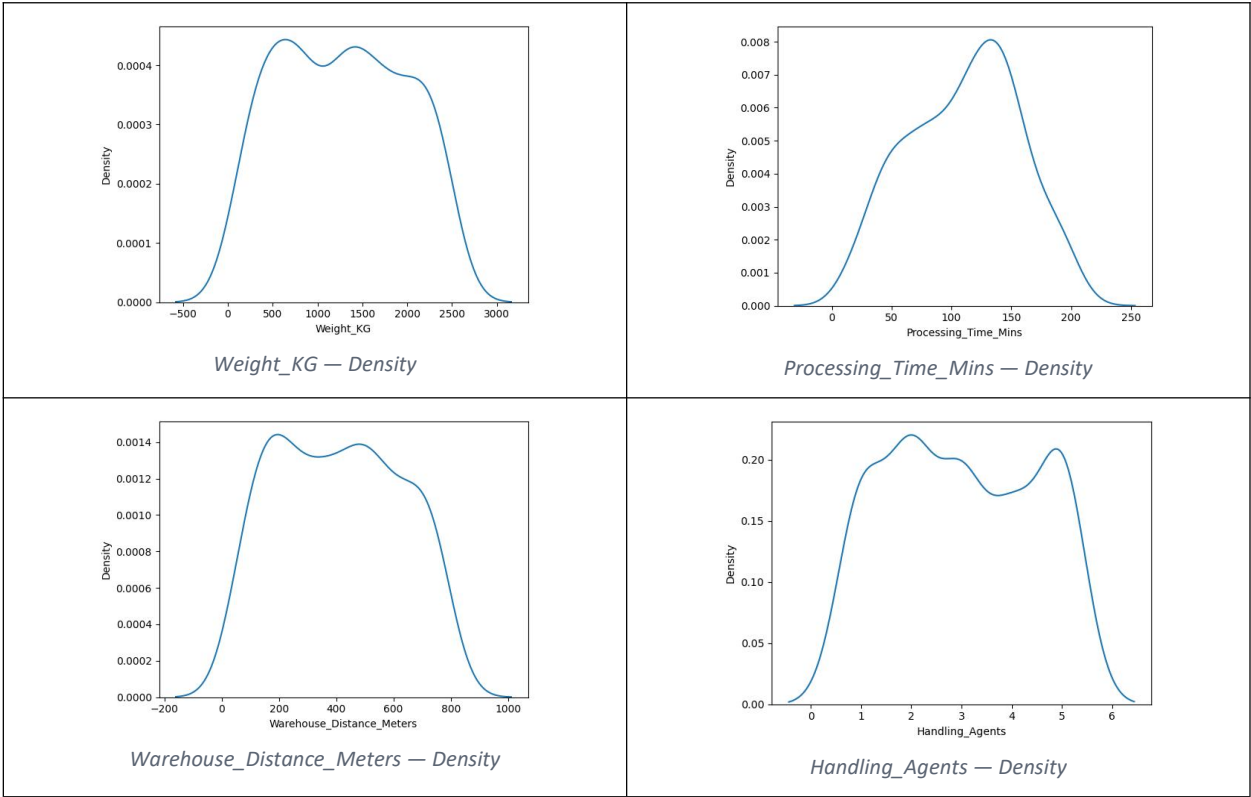
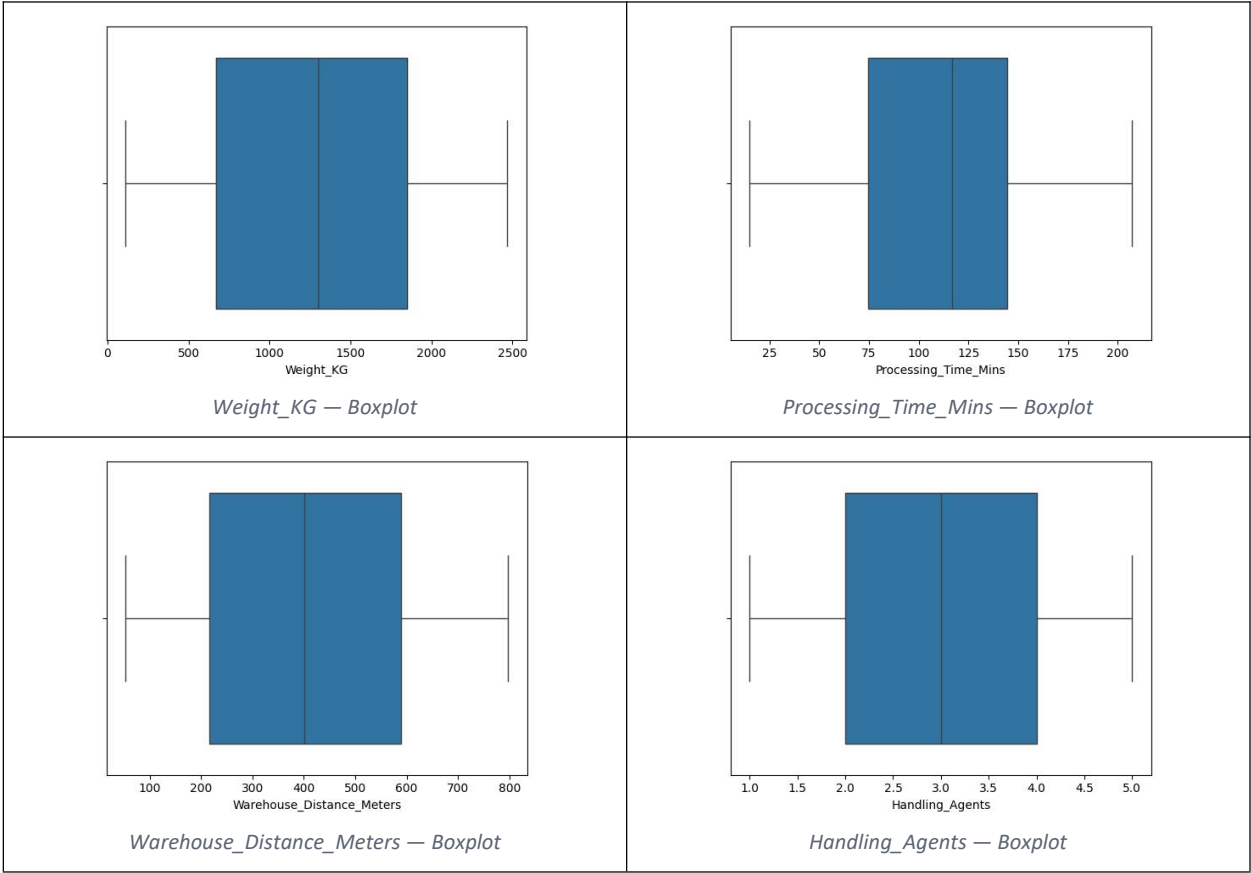
Missing Weight_KG values were imputed using the column median (a robust choice given the roughly symmetric distribution, skew ≈ 0.05), and comma separators were stripped before casting the column back to a numeric (float) type. A duplicate check confirmed zero duplicate records, so no rows were removed.

Post-cleaning, all 250 records were complete across all seven columns with no remaining nulls or duplicates.

4. Exploratory Data Analysis

4.1 Univariate Analysis — Distribution & Spread

Boxplots and kernel density estimates (KDE) were used to examine the shape, spread, and outlier profile of each numerical variable. All four numerical variables showed near-zero skewness (between -0.13 and 0.07), indicating approximately symmetric distributions with no significant outliers.



4.2 Bivariate Analysis — Numerical Relationships

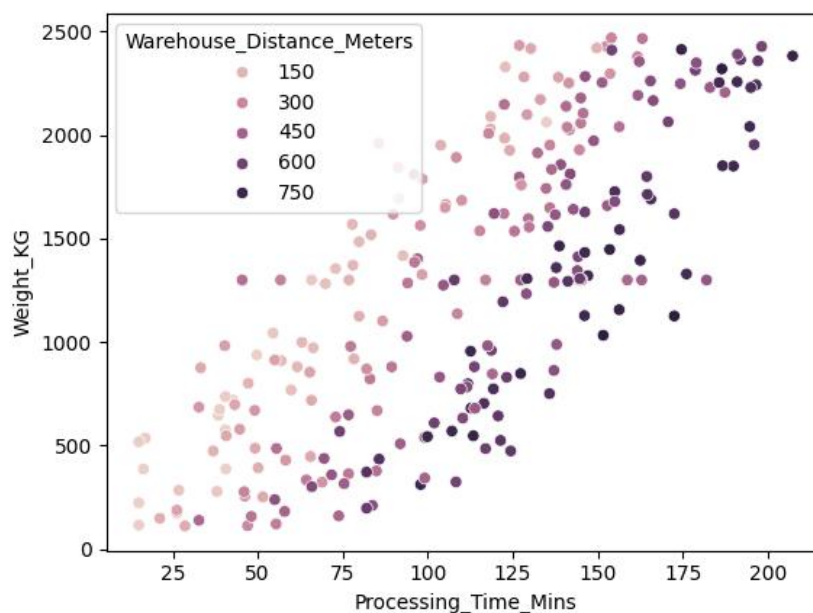


Figure: Processing Time vs. Weight, colored by Warehouse Distance

Observation

Some heavier shipments still show short processing times, which appears to be explained by shorter warehouse-to-gate distances — distance, not weight alone, drives a meaningful share of processing time.

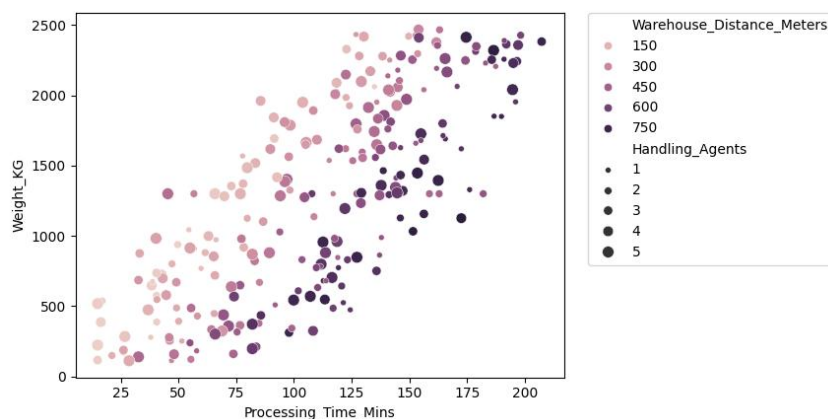


Figure: Processing Time vs. Weight, colored by Distance, sized by Handling Agents

Observation

Shipments with lower weight and shorter distance but staffed with more handling agents (5) tend to clear faster — suggesting agent staffing can partially offset shipment characteristics that would otherwise slow processing.

4.3 Bivariate Analysis — Numerical vs. Categorical

Processing time, weight, distance, and agent count were each compared against Priority_Level to test whether the priority tagging system operates as intended.

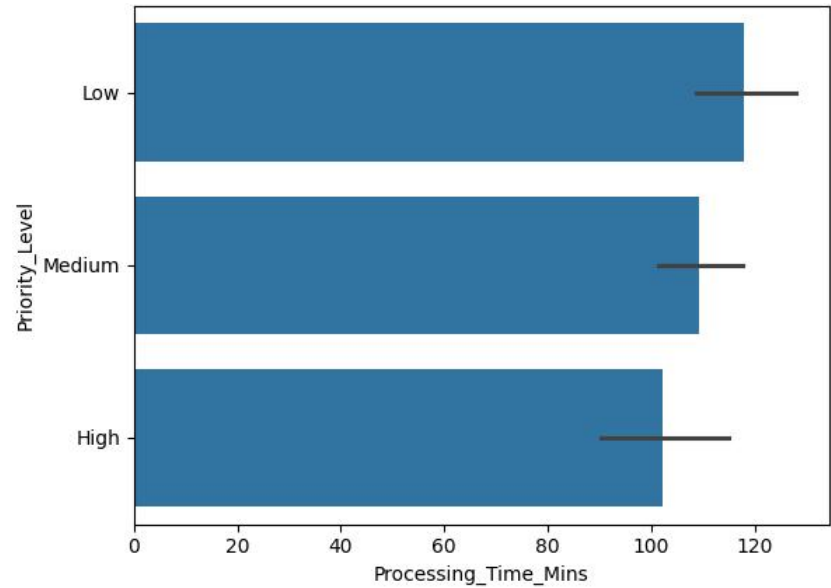


Figure: Processing Time by Priority Level

Processing Time vs. Priority

Strongly related — High-priority shipments show markedly lower processing time, confirming that the priority pipeline functions as intended.

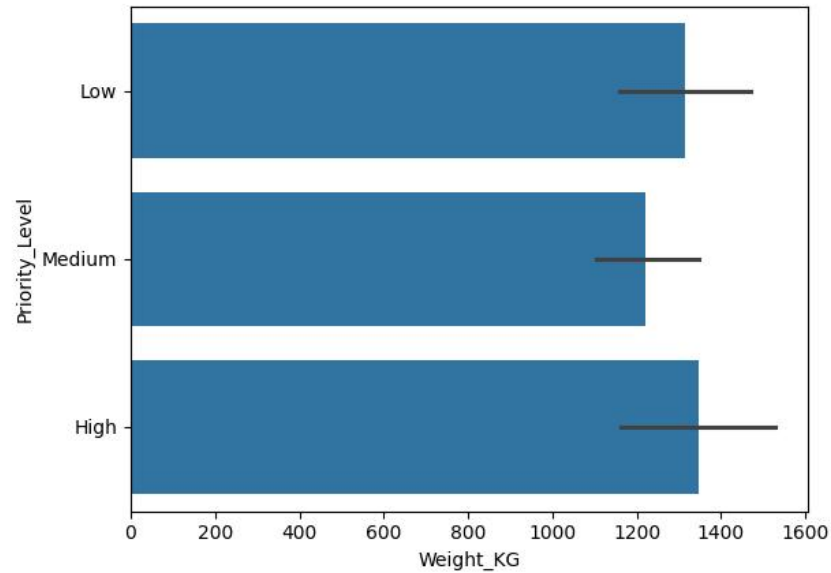


Figure: Weight by Priority Level

Weight vs. Priority

No meaningful relationship — heavy and light shipments are distributed evenly across all priority tiers.

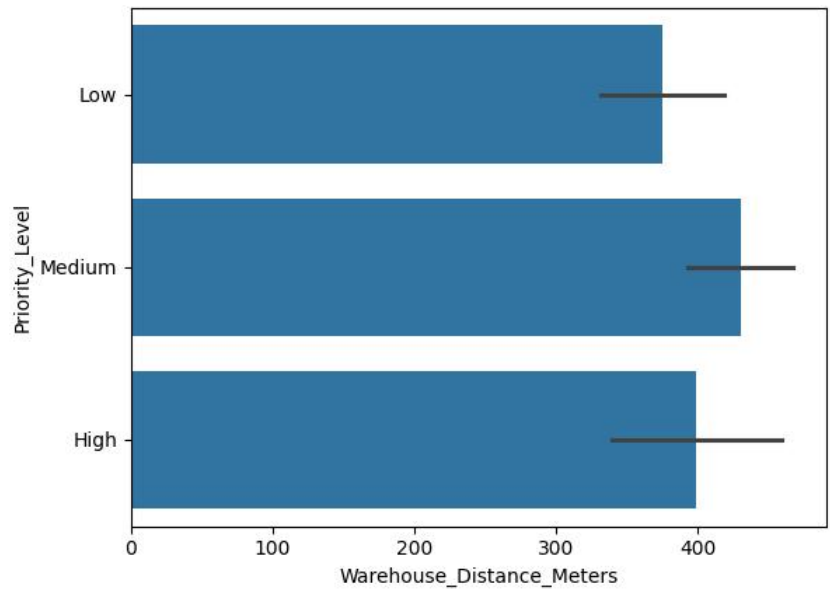


Figure: Warehouse Distance by Priority Level

Distance vs. Priority

No relationship — physical distance to the gate is independent of the priority tag assigned to a shipment.

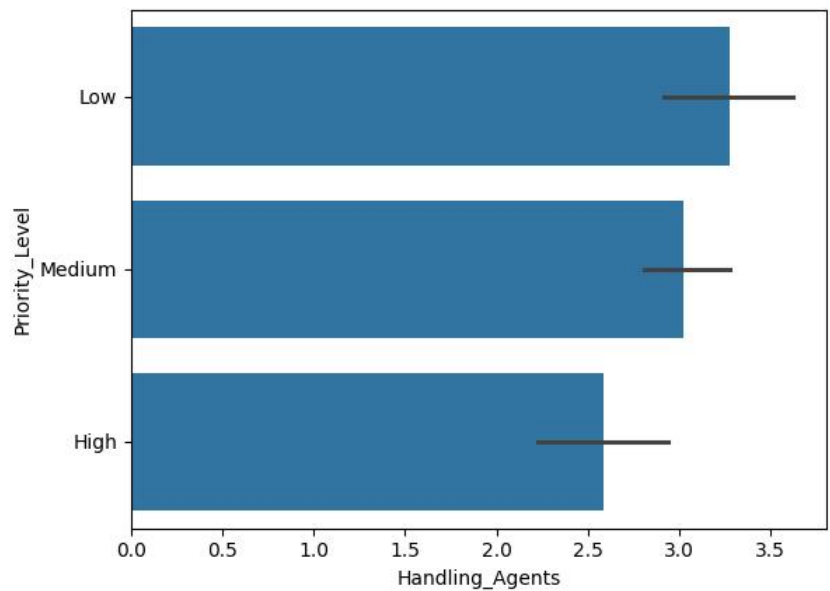


Figure: Handling Agents by Priority Level

Agents vs. Priority

Inverse relationship — fewer agents, on average, are assigned to High-priority shipments. Their processing speed advantage appears to come from operational efficiency or routing rather than sheer staffing volume.

4.4 Categorical × Categorical — Priority vs. Destination Zone

A cross-tabulation of Priority_Level against Destination_Zone was used to check for imbalance in how priority tags are distributed across zones.

Priority Level	Zone_A	Zone_B	Zone_C	Zone_D
High	23	11	7	12
Low	31	12	18	11
Medium	54	28	16	27

Zone_A carries the largest share of shipments across every priority level, consistent with it being the busiest destination zone in the dataset, but the relative mix of High/Medium/Low priority within each zone is broadly proportional.

5. Feature Engineering

Before modeling, the cleaned dataset was transformed into a fully numerical, model-ready format through the following steps:

- Ordinal encoding: Priority_Level was mapped to an ordered numeric scale (Low = 0, Medium = 1, High = 2), preserving its inherent ranking.
- One-hot encoding: Destination_Zone was expanded into three dummy indicator columns (Zone_B, Zone_C, Zone_D), with Zone_A used as the reference category (drop_first = True).
- Standardization: Handling_Agents, Warehouse_Distance_Meters, and Weight_KG were scaled to zero mean and unit variance using StandardScaler, putting all numerical features on a comparable scale.
- Identifier removal: Shipment_ID was dropped, as it carries no predictive signal about processing time.

The resulting feature set contained seven predictors — Weight_KG, Priority_Level, Warehouse_Distance_Meters, Handling_Agents, and three destination-zone dummy variables — used to predict Processing_Time_Mins.

6. Predictive Modeling

6.1 Model Setup

A Linear Regression model was selected as a first, interpretable baseline for forecasting Processing_Time_Mins. The dataset was split 80/20 into training and test sets (random_state = 42) to allow an honest evaluation of generalization performance.

6.2 Performance

Metric	Value	Interpretation
R ² Score	0.9342	93.4% of the variance in processing time is explained by the model
RMSE	10.78 minutes	Typical prediction error of roughly ± 11 minutes
Base Intercept	121.70 minutes	Predicted processing time when all features are at their reference/mean values

An R² of 0.93 indicates the linear model captures the great majority of the systematic variation in processing time, with only a small residual unexplained by the available features.

6.3 Feature Importance (Model Coefficients)

Because numerical features were standardized, the regression coefficients are directly comparable and indicate each feature's relative influence on processing time (in minutes, per one standard-deviation change).

Feature	Coefficient	Direction	Effect
Weight_KG	+31.11	Increases	Heavier shipments take substantially longer to process
Warehouse_Distance_Meters	+26.88	Increases	Greater warehouse-to-gate distance adds processing time
Priority_Level	-12.21	Decreases	Higher priority shipments are processed faster
Handling_Agents	-7.23	Decreases	More assigned agents modestly reduces processing time
Destination_Zone_Zone_C	-4.11	Decreases	Zone C shipments are slightly faster than the Zone A baseline
Destination_Zone_Zone_B	+3.00	Increases	Zone B shipments are slightly slower than the Zone A baseline
Destination_Zone_Zone_D	-1.23	Decreases	Zone D has a marginal, near-negligible effect

Key driver

Weight and warehouse distance are the two strongest drivers of processing time, followed by priority level. Destination zone contributes only a minor, second-order effect once the other factors are accounted for.

6.4 Actual vs. Predicted Performance

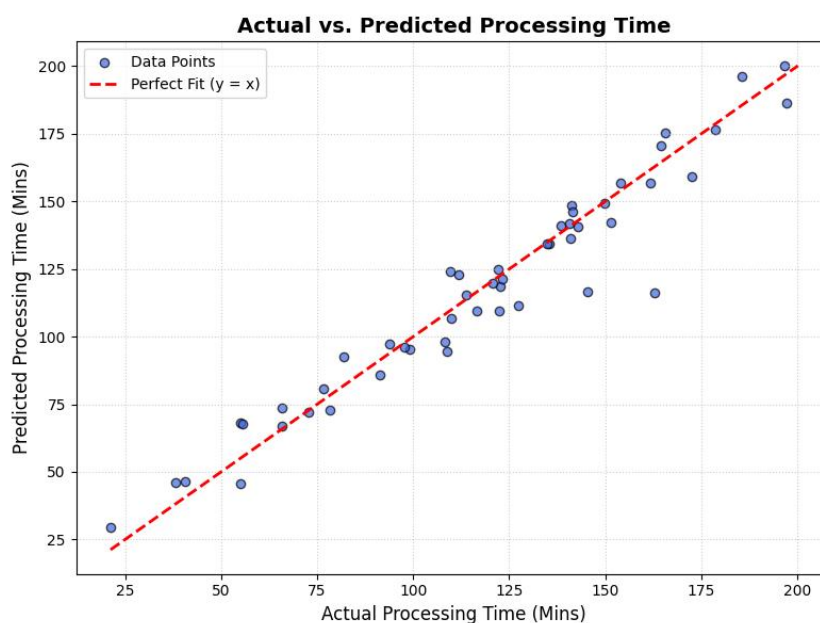


Figure: Actual vs. Predicted Processing Time (test set)

Predicted values track closely with actual processing times along the reference $y = x$ line, with only modest scatter — visually confirming the strong R^2 score and indicating no systematic bias (over- or under-prediction) across the range of the target variable.

7. Conclusions & Recommendations

7.1 Key Findings

- Shipment weight and warehouse-to-gate distance are the dominant drivers of processing time, together outweighing every other factor in the model.
- The priority tagging system works as designed — higher-priority shipments are processed measurably faster, independent of weight or distance.
- Handling agent count has a modest but real speed benefit, and its effect is distinct from — not explained by — priority level.
- Destination zone has only a marginal effect once weight, distance, priority, and staffing are accounted for.
- A simple linear model already explains 93.4% of the variance in processing time, indicating the relationship between these operational features and processing time is largely linear.

7.2 Operational Recommendations

- Use predicted processing time (from this model) to proactively flag heavy, long-distance shipments for additional staffing before bottlenecks occur.

- Investigate whether Zone B's slightly elevated processing time reflects a fixable logistics inefficiency (e.g., routing, dock congestion).
- Consider a controlled test of agent allocation on Low/Medium-priority shipments to quantify how much additional staffing could compress processing time.
- As more data accumulates, test non-linear models (e.g., Random Forest or Gradient Boosting) to check for interaction effects the linear model may be missing — for instance, between weight and distance.

7.3 Limitations

- The dataset is limited to 250 records; a larger sample would improve the reliability of the coefficient estimates and support more complex models.
- Weight_KG imputation via median fill (12 records, 4.8%) assumes missingness is unrelated to the true weight value; this should be revisited if a systematic cause of missingness is identified.
- The linear model assumes additive, linear effects; it may understate performance in edge cases with strong feature interactions.

End of report.